Circles

Class 3

Note for the teacher.

- **Aims.** Students name the minor and major arcs cut off by a chord and the angle each subtends at the centre. They discover, then explain, that the angle at the centre is double the angle at a point on the remaining part of the circle. They use this to show that angles in the same segment are equal and that the angle in a semicircle is 90°.
- **NCERT links.** Theorem 9 and its corollary keep their NCERT numbers. The questions include NCERT Exercise Set 5.6 and End-of-Chapter Questions 2, 6, 20 and 24.
- **Prior work.** Isosceles triangles (base angles equal), the angle sum of a triangle and the exterior angle property. Class 2: circumcircles.
- **Materials.** Two A4 sheets and the geometry box (protractor, compass, set square) for each student.
- **Method.** The *Solve first* problems P2 to P4 compute the same relation three times, in the three positions of the centre. Let students state the pattern before Theorem 9 is shown.

1 Arcs

Two points A and B on a circle cut it into two **arcs**. The smaller one is the **minor arc** and the larger one is the **major arc**. If AB is a diameter, the two arcs are equal; each is a **semicircle**.

FIG

The angle that an arc **subtends at the centre** is the angle swept as the radius turns from OA to OB along that arc. The minor arc AXB subtends $\angle AOB$. The major arc AYB subtends the reflex angle AOB, which is $360° - \angle AOB$.

Solve first.

P1. In a circle with centre O, $\angle AOB = 100°$.

(a) Which arc subtends 100° at the centre: the minor arc or the major arc?

(b) What angle does the other arc subtend at the centre?

(c) What angle does a semicircle subtend at the centre?

Answers.

P1. (a) The minor arc. (b) The major arc subtends the reflex angle $360° - 100° = 260°$. (c) 180°: a straight angle.

2 The angle at the centre and the angle on the circle

The angle that arc AB **subtends at a point** C on the rest of the circle is $\angle ACB$. In P2 to P4, O is the centre and CD is a diameter. Draw a rough diagram for each and mark every angle you find.

Solve first.

P2. A is a point on the circle, and $\angle OCA = 25°$.

(a) Why is $\angle OAC = 25°$?

(b) $\angle AOD$ is an exterior angle of $\triangle OAC$. Find it.

(c) Compare $\angle AOD$ with $\angle ACD$.

P3. A and B lie on opposite sides of the diameter CD, with $\angle OCA = 20°$ and $\angle OCB = 30°$.

(a) Use P2 to find $\angle AOD$ and $\angle BOD$.

(b) Find $\angle AOB$ and $\angle ACB$.

(c) Compare $\angle AOB$ with $\angle ACB$.

P4. A and B lie on the same side of the diameter CD, with $\angle OCA = 40°$ and $\angle OCB = 15°$.

(a) Find $\angle AOD$ and $\angle BOD$.

(b) Find $\angle AOB$ and $\angle ACB$.

(c) Compare $\angle AOB$ with $\angle ACB$. What pattern do P2, P3 and P4 show?

Answers.

P2.

(a) $OA = OC$ (radii), so the angles opposite them are equal.

(b) $\angle AOD = \angle OAC + \angle OCA = 25° + 25° = 50°$ (exterior angle).

(c) $\angle AOD = 50° = 2 \times 25° = 2\angle ACD$.

FIG

P3.

(a) As in P2: $\angle AOD = 2 \times 20° = 40°$ and $\angle BOD = 2 \times 30° = 60°$.

(b) $\angle AOB = 40° + 60° = 100°$ and $\angle ACB = 20° + 30° = 50°$.

(c) $\angle AOB = 2\angle ACB$.

FIG

P4.

(a) As in P2: $\angle AOD = 2 \times 40° = 80°$ and $\angle BOD = 2 \times 15° = 30°$.

(b) $\angle AOB = 80° - 30° = 50°$ and $\angle ACB = 40° - 15° = 25°$.

(c) $\angle AOB = 2\angle ACB$. In all three problems the angle at the centre is double the angle at C.

FIG

Note for the teacher. P2 is the key step: P3 adds two copies of it and P4 subtracts one from the other. These are the three positions of the centre: on one arm of the angle, inside the angle and outside it. NCERT treats the second and third cases.

Theorem 9 (NCERT Theorem 9). *The angle subtended by an arc at the centre of the circle is double the angle subtended by the arc at any point on the circle outside the arc.*

FIG

Why it works.

Given: arc AB of a circle with centre O; C is a point on the circle outside the arc.
To show: $\angle AOB = 2\angle ACB$.

1. Draw the diameter CD. $OA = OC$ (radii), so $\angle OAC = \angle OCA$. The exterior angle of $\triangle OAC$ is $\angle AOD = \angle OAC + \angle OCA = 2\angle OCA$.
2. In the same way, $\angle BOD = 2\angle OCB$.
3. If O lies inside $\angle ACB$, add: $\angle AOB = \angle AOD + \angle BOD = 2(\angle OCA + \angle OCB) = 2\angle ACB$.
4. If O lies outside $\angle ACB$, subtract: $\angle AOB = \angle AOD - \angle BOD = 2(\angle OCA - \angle OCB) = 2\angle ACB$.

The same steps work when the arc is a major arc; then $\angle AOB$ is a reflex angle.

Example 1. In a circle with centre O, $\angle AOB = 130°$, and C is a point on the major arc AB. Find $\angle ACB$ and $\angle OAB$.

FIG

Given: $\angle AOB = 130°$; C is on the major arc.
To find: $\angle ACB$ and $\angle OAB$.

1. C is outside the minor arc AB, which subtends $130°$ at O. So $\angle ACB = \dfrac{130°}{2} = 65°$ (Theorem 9).
2. $OA = OB$ (radii), so $\angle OAB = \angle OBA = \dfrac{180° - 130°}{2} = 25°$.

Answer: $\angle ACB = 65°$ and $\angle OAB = 25°$.

3 Two consequences

Solve first.

P5. In a circle with centre O, $\angle AOB = 80°$. C, D and E are three points on the major arc AB.

(a) Find $\angle ACB$, $\angle ADB$ and $\angle AEB$.
(b) Does the answer depend on where C is on the major arc?

P6. AB is a diameter of a circle with centre O, and C is any other point on the circle.

(a) What angle does the semicircle not containing C subtend at O?
(b) Find $\angle ACB$.

Answers.

P5. (a) Each is $\frac{80^\circ}{2} = 40^\circ$ (Theorem 9). (b) No. Every point of the major arc gives the same angle.

P6. (a) 180°. (b) $\angle ACB = \frac{180^\circ}{2} = 90^\circ$.

The part of the circle on one side of a chord, together with the chord, is a **segment**. P5 shows that *angles in the same segment are equal*: each is half the same angle at the centre.

Corollary of Theorem 9 (NCERT Corollary). *The angle subtended by a diameter at any point on the circle is* 90°.

FIG

Why it works.

Given: AB is a diameter of a circle with centre O; C is any other point on the circle.
To show: $\angle ACB = 90^\circ$.

1. The semicircle not containing C subtends a straight angle, 180°, at O.
2. By Theorem 9, $\angle ACB = \frac{180^\circ}{2} = 90^\circ$.

Things to Remember

The angle at the centre is double the angle on the circle, for the same arc.
Angles in the same segment are equal.
The angle in a semicircle is a right angle.

Solve first.

P7. Draw a circle and a chord AB. Mark a point C on the major arc, a point P inside the circle and a point Q outside it, all on the same side of AB. Measure $\angle ACB$, $\angle APB$ and $\angle AQB$. Which is the largest? Which is the smallest?

Answers.

P7. $\angle APB > \angle ACB > \angle AQB$: a point inside the circle gives a larger angle, and a point outside gives a smaller one. Only points on the arc give the angle $\angle ACB$. Class 4 uses this fact.

Example 2. AB is a diameter of a circle with centre O, and C is a point on the circle with $\angle CAB = 35^\circ$. Find $\angle ABC$.

FIG

Given: AB is a diameter; $\angle CAB = 35^\circ$.
To find: $\angle ABC$.

1. $\angle ACB = 90°$ (angle in a semicircle).
2. $\angle ABC = 180° - 90° - 35° = 55°$ (angle sum of a triangle).

Answer: $\angle ABC = 55°$.

Draw a rough figure for every question in Exercise 3A that has no figure.

Exercise 3A

1. An arc of a circle subtends an angle of $70°$ at the centre. Find the angle it subtends at a point on the rest of the circle.
2. In a circle with centre O, C is a point on the major arc AB and $\angle ACB = 40°$. Find $\angle AOB$ and $\angle OAB$.
3. In a circle with centre O and radius 12 cm, $\angle AOB = 60°$. Find the length of the chord AB, and the angle that AB subtends at a point on the major arc.
4. In a circle with centre O, $\angle AOB = 100°$, and D is a point on the *minor* arc AB. Ravi writes $\angle ADB = 50°$. Find his mistake, and find $\angle ADB$.
5. AB is a diameter of a circle and C is any other point on the circle. Find $\angle ACB$ and explain your reasoning without using Theorem 9: join OC and use two isosceles triangles.
6. Find x in each figure. O is the centre.

FIG

7. A carpenter wants to check that a wooden piece has a true semicircular edge. She places the corner of a set square on the curved edge so that its two edges pass through the ends of the straight edge. She repeats this at several places. Explain the test.
8. $MNOP$ is a quadrilateral inscribed in a circle, and MN is a diameter. What can you say about $\angle MOP$ and $\angle MNP$? Explain.
9. A and B are points on a circle.
   (a) Can there be points X and Y on the circle, on the same side of AB, with $\angle AXB \neq \angle AYB$?
   (b) If X and Y are on the circle and $\angle AXB = \angle AYB$, must X and Y be on the same side of AB?
   (c) If $\angle AXB = \angle AYB$, and X and Y are not on the circle, does the circle through A, B and X also pass through Y?
10. In Class 2 we saw by construction that the circumcentre of a right-angled triangle is the midpoint of its hypotenuse. Prove it. Let $\triangle PQR$ have $\angle P = 90°$ and circumcentre O, and use Theorem 9 for the arc QR that does not contain P.
11. A chord of a circle is equal to the radius. Find the angle it subtends at a point on the major arc.

Note for the teacher. Question 4 must be discussed in class: students apply "half" without checking which arc the point lies on. Question 9 prepares Class 4. If time is short, set Questions 8, 10 and 11 as homework.

Answers.

1. $\dfrac{70^\circ}{2} = 35^\circ$ (Theorem 9).

2.

1. $\angle AOB = 2\angle ACB = 80^\circ$ (Theorem 9). FIG
2. $OA = OB$, so $\angle OAB = \dfrac{180^\circ - 80^\circ}{2} = 50^\circ$.

Answer: $\angle AOB = 80^\circ$, $\angle OAB = 50^\circ$.

3.

1. $OA = OB = 12$ cm and $\angle AOB = 60^\circ$. The base angles are each 60°, so $\triangle OAB$ is equilateral and $AB = 12$ cm. FIG
2. At a point on the major arc, AB subtends $\dfrac{60^\circ}{2} = 30^\circ$ (Theorem 9).

Answer: $AB = 12$ cm; 30°.

4.

D lies *on* the minor arc, so it is outside the major arc. The major arc subtends the reflex angle $360^\circ - 100^\circ = 260^\circ$ at O. So $\angle ADB = \dfrac{260^\circ}{2} = 130^\circ$ (Theorem 9). FIG

Answer: 130°. Ravi halved the angle of the wrong arc.

5.

1. $OA = OC = OB$ (radii). Let $\angle OAC = \angle OCA = a$ and $\angle OBC = \angle OCB = b$. FIG
2. Angle sum of $\triangle ABC$: $a + (a + b) + b = 180^\circ$, so $2(a + b) = 180^\circ$.
3. $\angle ACB = a + b = 90^\circ$.

This is NCERT End-of-Chapter Question 24.

6.

(a) $\angle ADB$ and $\angle ACB$ are in the same segment: $x = 30^\circ$. FIG

(b) $x = 2 \times 48^\circ = 96^\circ$ (Theorem 9).

(c) $\angle PRQ = 90^\circ$ (angle in a semicircle), so $x = 180^\circ - 90^\circ - 28^\circ = 62^\circ$.

7. The corner of the set square is a right angle. If the edge is a true semicircle, the angle in a semicircle is 90° at *every* point of the curve, so the corner touches the curve at every position. If the corner leaves a gap at some place, the edge is not a true semicircle there.

8.

$\angle MOP$ and $\angle MNP$ both stand on the arc MP and lie in the same segment, so $\angle MOP = \angle MNP$. Also $\angle MON = \angle MPN = 90^\circ$ (angles in a semicircle). FIG

9.

(a) No. Points on the same side of AB lie in the same segment, so the angles are equal.

(b) No. If AB is a diameter, every point of the circle on either side gives $90°$ (angle in a semicircle).

(c) Yes, if X and Y are on the same side of AB. This is Theorem 10, proved in Class 4.

10.

1. $\angle QPR = 90°$, and P is outside the arc QR that does not contain P. By Theorem 9, this arc subtends $2 \times 90° = 180°$ at O.
2. So $\angle QOR = 180°$: Q, O and R lie on one straight line, and QR is a diameter.
3. $OQ = OR$ (radii), so O is the midpoint of the hypotenuse QR.

FIG

11. The chord and the two radii form an equilateral triangle, so the chord subtends $60°$ at the centre and $\frac{60°}{2} = 30°$ at a point on the major arc.